

DDCA PROJECT PRESENTATION

Design and Simulation of Adder and Subtractor Using Logisim

Batch No: 5

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🎯 Introduction & Core Objectives

🔧 Digital Fundamentals

Adders and subtractors are essential **combinational circuits** forming the core arithmetic unit of CPUs and microcontrollers.

Combinational Logic: Outputs strictly depend on current inputs with no memory or feedback paths.

Key Concept: Using 2's complement system converts binary subtraction into simple binary addition, reducing circuit gate count and energy consumption.

📋 Project Objectives

Simulate and analyze fundamental digital circuits using **Logisim** software to verify logical behavior and propagation delays.

- ✓ Construct Half & Full Adder subcircuits
- ✓ Construct Half & Full Subtractor subcircuits
- ✓ Build 4-Bit Adder-Subtractor with Mode selector
- ✓ Verify outputs against theoretical truth tables

Adder & Subtractor Boolean Logic

+ Binary Adder Equations

Half Adder (2 Inputs A, B):

$$\text{Sum} = A \oplus B \quad | \quad \text{Carry} = A \cdot B$$

Full Adder (3 Inputs A, B, C_{in}):

$$\text{Sum} = A \oplus B \oplus C_{in}$$

$$C_{out} = (A \cdot B) + (C_{in} \cdot (A \oplus B))$$

Full Adder uses 2 XOR, 2 AND, and 1 OR gate.

— Binary Subtractor Equations

Half Subtractor (2 Inputs A, B):

$$\text{Diff} = A \oplus B \quad | \quad \text{Borrow} = \bar{A} \cdot B$$

Full Subtractor (3 Inputs A, B, B_{in}):

$$\text{Diff} = A \oplus B \oplus B_{in}$$

$$B_{out} = (\bar{A} \cdot B) + (B_{in} \cdot A \oplus B)$$

Full Subtractor incorporates inverter on minuend bit A.

4-Bit Combined Unit Architecture

-  **Mode Pin (M):** Controls operational mode. $M=0$ configures addition; $M=1$ configures 2's complement subtraction.
-  **XOR Controlled Inverter:** Acts as conditional buffer ($B \oplus 0 = B$) or bitwise NOT gate ($B \oplus 1 = \bar{B}$).
-  **Overflow Detection (V):** Calculated via XOR between carry-in to MSB and carry-out from MSB: $V = C_3 \oplus C_4$.

M = 0 (Addition)

Passes B directly, sets $C_0 = 0$.

Operation = $A + B$

M = 1 (Subtraction)

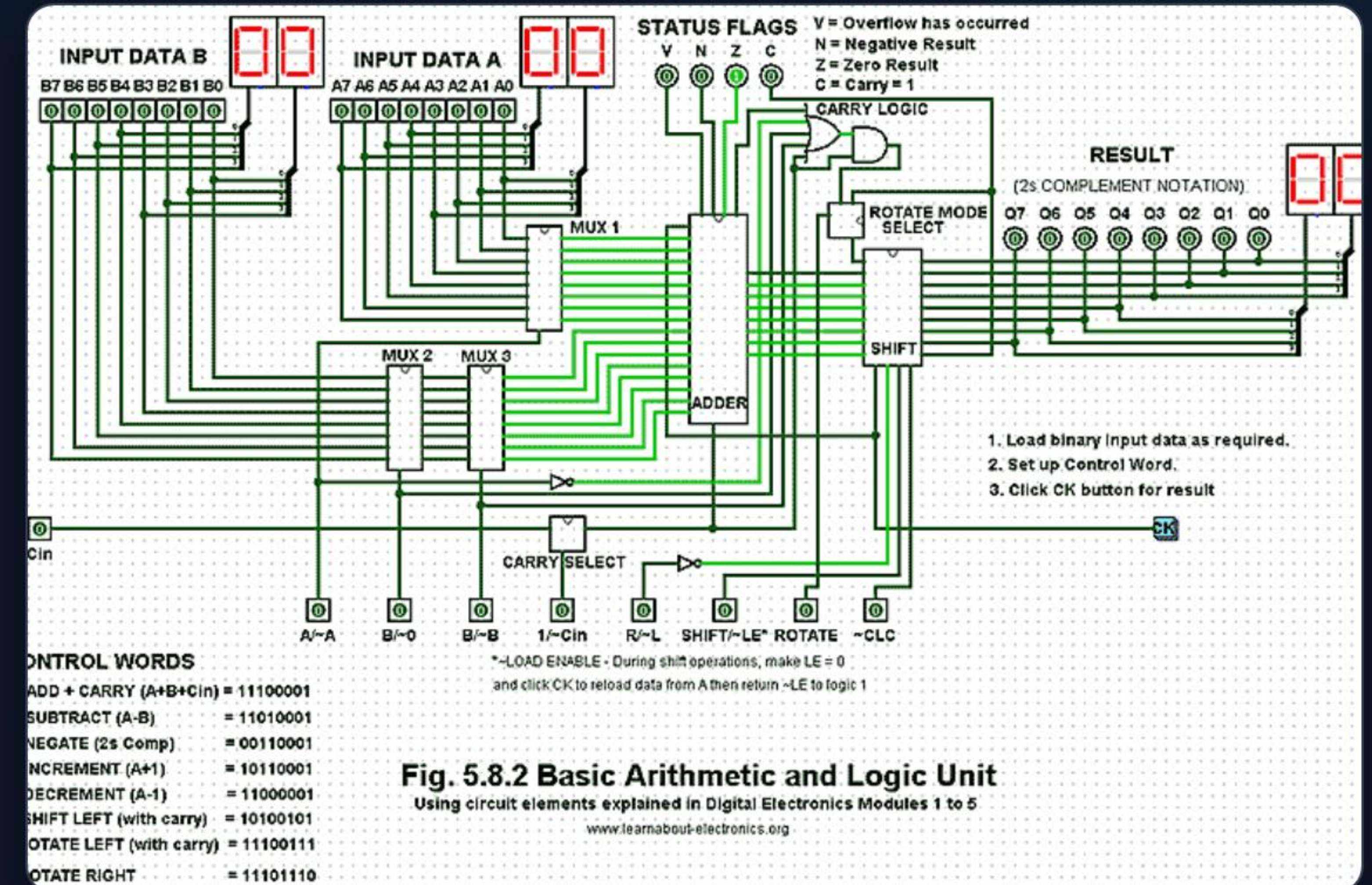
Inverts B, sets initial carry $C_0 = 1$.

Operation = $A + \bar{B} + 1 = A - B$

4-Bit Adder Subtractor Diagram

Logisim Simulation & Utility

-  **Modular Hierarchical Design:** Created Full Adder subcircuits first, then instantiated them 4 times in the top-level schematic for clean modular design.
-  **Propagation Delay Analysis:** Observed ripple carry delay across 4 stages, highlighting latency trade-offs compared to Carry Lookahead Adders (CLA).
-  **ALU Core Building Block:** Forms the core mathematical processor inside CPUs for binary calculation, address generation, and flag generation.

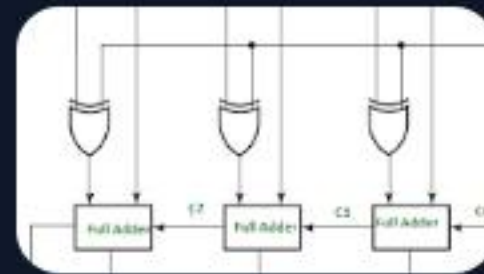


Thank You!

Questions & Answers

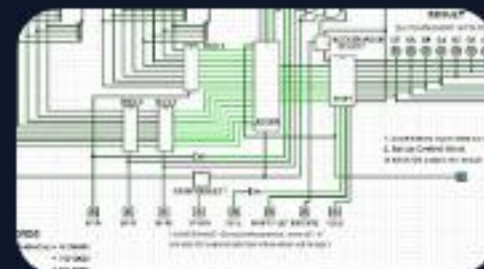
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